

# On the decomposition of a strong epimorphism into regular epimorphisms

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July 13, 2026. CT2026

This talk is based on a preprint [arXiv: 2604.05744](https://arxiv.org/abs/2604.05744).

This work was supported by JSPS KAKENHI Grant Numbers JP24KJ1462, JP25KJ1474.

1 The decomposition number

2 Minimum vs. canonical

# Regular-epi decomposition

Recall: the fundamental theorem of homomorphisms

For  $A \xrightarrow{f} B$ ,

$$A / \text{Ker}(f) \quad \underbrace{A / \text{Ker}(f)}_{\text{the coeq. of the kernel pair}} \cong \text{Im}(f)$$

the smallest s

Equivalently, the unique morphism  $f_1$  becomes mono:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow \text{reg. epi} & \nearrow f_1 \\ & \text{Coim } f := A / \text{Ker } f & \end{array}$$

The theorem holds in any category of equational algebras.

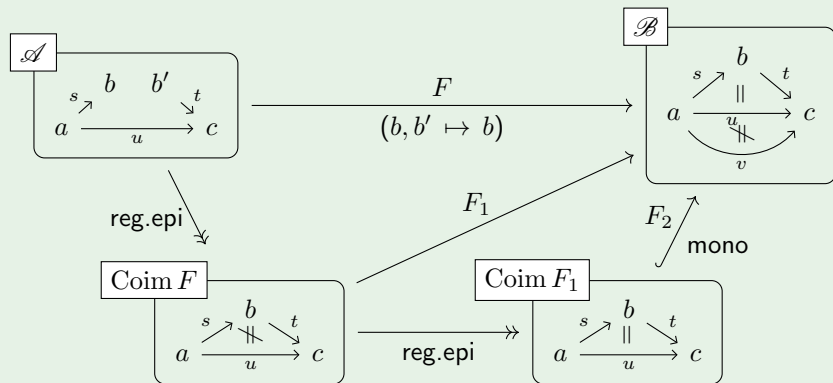
## Question

Does the theorem hold in any category?  $\rightsquigarrow$  No.

# The case of Cat

## Counterexample

In Cat,



# The decomposition number

## Definition [GU71; MS82; Bör91; KN26]

- 1 The (canonical) decomposition number  $\sigma(f) :=$  the smallest  $\alpha$  s.t. " $f_\alpha$ " becomes mono.
- 2 For a category  $\mathcal{C}$ ,  $\sigma(\mathcal{C}) :=$  the smallest  $\beta$  s.t.  $\forall f$  in  $\mathcal{C}$ ,  $\sigma(f) < \beta$ .

## Example

- 1  $\sigma(\mathbf{Set}) = \sigma(\mathbf{Grp}) = 2$ . (the fundamental thm. of homomorphisms)
- 2  $\sigma(\mathbf{Pos}) = 2$ .
- 3  $\sigma(\mathbf{Field}) = 1$ .
- 4  $\sigma(\mathbf{Cat}) = 3$  [BBP99].

## Theorem [GU71]

$\mathcal{C}$ : locally  $\kappa$ -presentable  $\rightsquigarrow \sigma(\mathcal{C}) \leq \kappa + 1$ .

However, this only provides a rough upper bound  $\sigma(\mathbf{Cat}) \leq \omega + 1$ .

# Giving a tight upper bound

$\mathcal{C}$ : locally presentable

**Task:** Find a stricter upper bound for  $\sigma(\mathcal{C})$ .

**Strategy:**

- 1 Construct a logical theory  $\mathbb{T}$  s.t.  $\mathbf{Mod} \mathbb{T} \simeq \mathcal{C}$ .
- 2 Looking closely at  $\mathbb{T}$ , we would obtain an upper bound.

## Theorem (informal) [KN26]

This strategy works well if  $\mathbb{T}$  is a *partial Horn theory* [PV07] or a *generalized algebraic theory* [Car86].

## Application

$$\begin{aligned}\sigma(n\text{-Cat}) &\leq n + 2, & \sigma(\mathbf{StrMonCat}) &\leq 3 \\ \sigma(\mathbf{MultiCat}) &\leq 3, & \sigma(\mathbf{DblCat}) &\leq 4.\end{aligned}$$

In fact, these are equalities!

1 The decomposition number

2 Minimum vs. canonical

# Minimum vs. canonical

## Recall

$\sigma(f) :=$  the “length” of the **canonical** decomp. of  $f$ . (taking the coimage)

How does this relate to **arbitrary** decompositions?

## Definition [KN26]

The **minimum decomposition number**  $\delta(f) :=$  the shortest possible length of decompositions of  $f$ .

## Theorem [KN26]

In a category with kernel pairs and colimits,

$$\sigma(f) = \delta(f).$$

That is, “**canonical = minimum.**” (This can be verified easily.)

## Further generalization

We started from:

- regular epimorphisms, and
- the canonical decompositions under them.

→ These can be generalized to a class  $\mathbf{E}$  of morphisms that admits *locally orthogonal  $\mathbf{E}$ -factorizations* [Tho79; MT82; Tho83].

### Example

- 1  $\mathbf{E} := \{\text{strict localizations in } \mathbf{Cat}\}.$
- 2  $\mathbf{E} := \{\text{monotone quotient maps in } \mathbf{Top}\}.$
- 3  $\mathbf{E} := \{\text{strictly monadic functors in } \mathbf{Cat}_{\text{radj}}\}.$

### Theorem [KN26]

For a class  $\mathbf{E}$  of morphisms that admits locally orthogonal  $\mathbf{E}$ -factorizations,

- 1 If  $\mathbf{E} \subseteq \{\text{epis}\}$ , then  $\sigma(f) = \delta(f)$ .
- 2 If  $\mathbf{E} \not\subseteq \{\text{epis}\}$ , then  $\sigma(f) = \delta(f)$  but only for finite ordinals. (Non-trivial)

# Thank you!

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